# Optimizing Inventory Management

Data-driven replenishment strategies and impact forecast

# Objective & Success Metrics

## Goal

Implement **data-driven demand forecasting and replenishment** to **minimize stockouts and lost sales** while ensuring **optimal inventory levels** throughout the retail network.

## Objective

Enhance product availability (Fill Rate) while maintaining consistent inventory control across stores and SKUs.

## Business Challenges

- **Stockouts** → lost sales and diminished customer satisfaction
- **Overstock** → tied-up capital and increased holding expenses

## Key Performance Indicator

**Fill Rate (Service Level)**

→ The percentage of actual customer demand met without stockouts

## Supporting Metrics

- **Forecast Accuracy:** WAPE, MAPE, Bias
- **Availability:** Stockout Rate, Lost Sales (Units / ₹)
- **Inventory Management:** DOH, Overstock Rate, Inventory Turns

# Data & Methodology (ETL + Forecasting)

## Data Coverage (Store–SKU–Day)

- **Sales:** Units sold, revenue
- **Inventory:** Daily on-hand inventory snapshots
- **Purchase Orders:** Order quantities, realized lead times
- **Product Master:** Category, cost, shelf life, MOQ
- **Calendar:** Day of week, promotion flag, holiday flag

## ETL Methodology

- Build a **complete daily time series** for every store–SKU
- Standardize IDs, handle missing dates, deduplicate records
- Outlier detection on demand using IQR method
- Forward-fill inventory to ensure continuity for planning

## Forecasting Methodology

- **Granularity:** Store–SKU–Day
- **Horizon:** 28 days (4 weeks)
- **Models evaluated:**
    - Seasonal Naive (weekly baseline)
    - Regression model (selected)
- **Features used:** Day of week, promo flag, holiday flag
- **Model selection criteria:** Lower WAPE, reduced bias, better promo handling

# Data & Methodology (Replenishment Policy)

## Policy Type

**Store–SKU Order-Up-To Replenishment Policy**

## Key Inputs

- Average daily demand (last 8 weeks)
- Demand volatility (standard deviation)
- Supplier lead time (median, with fallback default)
- Category-specific service level targets

## Core Calculations

- **Safety Stock** = $z \times \sigma \times \sqrt{\text{LeadTime}}$
- **Reorder Point (ROP)** = AvgDailyDemand × LeadTime + Safety Stock
- **Order Quantity** = Raise inventory to **30 days of cover**

## Operational Constraints Applied

- Minimum Order Quantity (MOQ)
- Shelf-life protection for perishable SKUs
- Optional store capacity limits

# Key Insights

- Stockouts are **highly concentrated**, not evenly distributed
- A **small number of categories** drive most stockout days
- High-stockout categories show **higher demand volatility**
- Promotions materially uplift demand → under-forecasting risk if ignored
- A subset of stores contributes disproportionately to stockouts
- Overstock risk driven by **excess DOH (>45 days)**
- Targeted replenishment can reduce stockouts without inflating inventory

# Forecast Performance & Chosen Approach

## Evaluation Metrics

- **WAPE (Primary):** Volume-weighted forecast accuracy
- **MAPE (Diagnostic):** Interpretability check (ex-zero demand days)
- **Forecast Bias:** Directional risk of over- or under-forecasting

## Performance Results

- **Regression model outperforms Seasonal Naive on WAPE**
- **Bias closer to zero**, reducing systematic overstock and stockout risk
- **Consistent performance across categories**, including high-volume SKUs

## Why This Approach Was Chosen

- Better captures **weekly patterns + promo/holiday effects**
- More **stable demand signal** for replenishment decisions
- Lower risk of inventory imbalance due to reduced bias

## Validation Diagnostics Used

- Actual vs forecast **trend alignment**
- **Forecast error distribution** centred around zero
- **Category-level WAPE comparison** to ensure robustness

# Replenishment Plan: How It Works

**Policy Type**

**Store–SKU Order-Up-To Replenishment Policy**

**How the Plan Works (Step-by-Step)**

1. **Estimate demand** using recent 8-week average and volatility
2. **Account for supply uncertainty** using realized supplier lead times
3. **Protect service levels** via category-specific safety stock
4. **Trigger replenishment** when on-hand inventory falls below Reorder Point
5. **Order up to a target inventory level** (30 days of cover)

**Key Inputs**

- Avg daily demand (last 8 weeks)
- Demand volatility (standard deviation)
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**Core Calculations**

- **Safety Stock** = $z \times \sigma \times \sqrt{\text{LeadTime}}$
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- **Order Quantity** = Raise inventory to **30 days of cover**

# Replenishment Plan: Examples

## Example 1 — Stockout-Risk SKU

- Avg daily demand: **8 units**
- Lead time: **4 days**
- Service level target: **97%**
- Current on-hand inventory: **Below ROP**

### Action:

Trigger replenishment to restore inventory to **30-day cover**, preventing imminent stockout and lost sales.

## Example 2 — Overstock-Risk SKU

- Days of Inventory on Hand (DOH): **> 45 days**
- Demand covered well beyond near-term requirements

### Action:

Reduce or pause future orders

Transfer excess stock to high-demand locations

Apply discounts where appropriate to accelerate sell-through

# 30-Day Impact Estimate (Assumptions & Scenarios)

## Assumptions & Methodology

- **Stockout defined** when on-hand inventory < Reorder Point (ROP)
- **Lost sales proxy** = Avg daily demand × supplier lead time
- Impact evaluated over a **30-day forward horizon**
- Execution uncertainty modelled using **±20% variance**

## Estimated Business Impact

- **Significant reduction in stockout days** for high-risk store–SKU combinations
- **Material recovery of lost sales (₹)** driven by improved availability
- **Controlled increase in inventory value**, reflecting a deliberate service-level trade-off

## Scenarios Evaluated

- **Base Case:** Expected execution
- **Best Case (+20%):** Strong replenishment execution
- **Worst Case (–20%):** Partial execution / operational friction

# Risks & Limitations

## Key Risks & Limitations

- **Linear forecasting model** may not fully capture nonlinear demand patterns
- **Promotional uplift** assumed uniform across stores and SKUs
- **Lead-time variability** partially captured using historical medians
- **Lost sales** estimated using a proxy, not actual unmet customer demand

## Current Mitigations

- Conservative, category-specific service level targets
- Scenario-based impact estimation (base / best / worst cases)
- Ongoing category- and store-level performance monitoring

# Risks & Limitations

## Recommendation

- Roll out the replenishment policy in a **phased manner**, starting with:
- **Top stockout-driving categories**
- **High-impact store–SKU combinations**
  This enables quick value capture while limiting operational risk.

## Next Steps

- Incorporate **promo depth and intensity** into demand forecasts
- Explicitly model **lead-time uncertainty and variability**
- Enable **store-specific service level tuning**
- Automate the **weekly replenishment planning and execution loop**

## Expected Outcomes

- Higher Fill Rate (improved availability)
- Reduced lost sales
- Disciplined and sustainable inventory growth